ASSIGNMENT 8: Apply Regularization to Avoid Overfitting

1. Aim

To implement Ridge Regression and Lasso Regression and compare their performance with ordinary Linear Regression to understand the role of regularization in controlling overfitting.

2. Objectives

1. To understand overfitting in machine learning.
2. To study the concept of regularization.
3. To understand L1 and L2 penalties.
4. To implement Ridge Regression and Lasso Regression.
5. To compare different regression models.
6. To evaluate their performance using RMSE and R-squared.

3. Theory

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary patterns. An overfitted model may show good performance on training data but poor performance on unseen data.

Regularization is a technique used to control model complexity by adding a penalty to the model's objective function. This discourages unnecessarily large coefficients and can improve generalisation.

Ridge Regression

Ridge Regression uses L2 regularization. It adds a penalty based on the sum of the squared coefficients. Ridge generally reduces coefficient magnitudes without setting them exactly to zero.

Lasso Regression

Lasso Regression uses L1 regularization. It adds a penalty based on the sum of the absolute coefficients. Lasso can reduce some coefficients to zero, which may help select a smaller set of features.

The regularization strength is controlled by a parameter called alpha. A larger alpha generally applies a stronger penalty.

To compare the techniques fairly, the models should use the same training and testing data. Standardisation is also useful because regularization is affected by the scales of input features.

In this assignment, ordinary Linear Regression, Ridge Regression, and Lasso Regression are evaluated using the Diabetes dataset. The test RMSE and R-squared values are compared.

Regularization can reduce overfitting, but its effectiveness depends on the dataset and the selected parameter values.

4. Algorithm

1. Start the program.
2. Import the required libraries.
3. Load the Diabetes dataset.
4. Separate input features and target values.
5. Divide the data into training and testing subsets.
6. Create Linear Regression, Ridge Regression, and Lasso Regression models.
7. Standardise the input features.
8. Train each model using the same training data.
9. Predict the test values.
10. Calculate RMSE and R-squared.
11. Compare the results.
12. Stop the program.

5. Explanation of the Code

1. `load_diabetes()` loads a numerical dataset suitable for regression.
2. `train_test_split()` divides the data into training and testing subsets.
3. `StandardScaler()` standardises the input features.
4. `LinearRegression()` creates the ordinary regression model.
5. `Ridge(alpha=1.0)` creates a model using L2 regularization.
6. `Lasso(alpha=0.1)` creates a model using L1 regularization.
7. `make_pipeline()` combines standardisation and regression.
8. `model.fit()` trains each model.
9. `model.predict()` predicts the target values for the test data.
10. RMSE measures prediction errors.

11. R-squared evaluates how well the model explains variation in the target.

6. Applications

1. Predicting sales from multiple features.
2. Forecasting numerical business outcomes.
3. Analysing datasets with correlated features.
4. Reducing the influence of unnecessarily large coefficients.
5. Selecting useful features using Lasso Regression.
6. Developing regression models that generalise to new observations.

7. Advantages

1. Regularization helps control model complexity.
2. Ridge Regression reduces the magnitude of coefficients.
3. Lasso Regression can set less useful coefficients to zero.
4. Regularization may improve performance on unseen data.
5. The models can be compared using consistent evaluation metrics.

8. Limitations

1. The regularization strength must be chosen carefully.
2. Excessive regularization can cause underfitting.
3. Results depend on the characteristics of the dataset.
4. Regularization does not guarantee better test performance.
5. Lasso may select only one feature from a group of highly correlated features.

9. Conclusion

Linear Regression, Ridge Regression, and Lasso Regression were implemented using the Diabetes dataset. Their performance was compared using test RMSE and R-squared. The experiment demonstrated how L1 and L2 regularization can control model complexity and potentially improve generalisation.